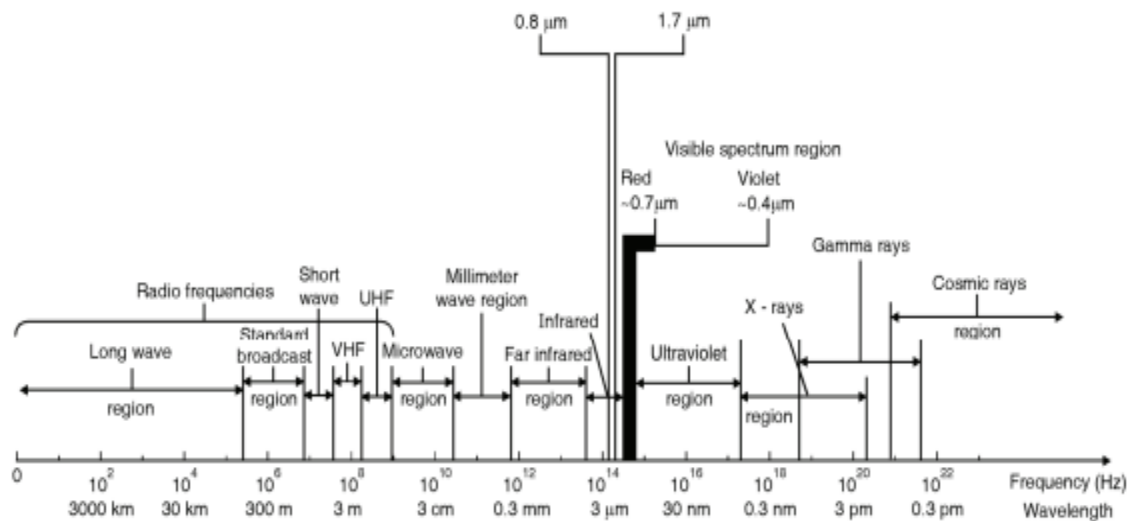


FIBER OPTIC COMMUNICATION (IN-SEM Notes) **(UNITS-1&2)**

Unit 1: OPTICAL FIBERS FOR TELECOMMUNICATIONS

Q. 1.1 Draw and explain an electromagnetic spectrum..... (4 Marks)

In the case of optical, information is transmitted in the form of light rays. The full range of wavelengths of light emitted from the hot body is called the 'spectrum'. This covers both visible and invisible regions. The electromagnetic spectrum covers the range of wavelengths from 10^4 meters to $1 \text{ \AA}$. ($1 \text{ \AA} = 10^{-8} \text{ cm}$). This electromagnetic spectrum is shown in Fig. 1.



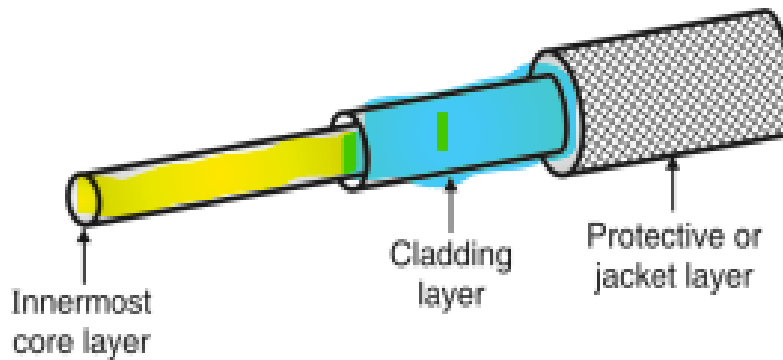
The frequency range and applications of each band are shown in the following table.

Designation	Frequency Range	Free-space Wavelength Range	Typical Applications
ELF (Extremely Low Frequency)	30–300 Hz	10,000–1000 km	Power line communications
VF (Voice Frequency)	300–3000 Hz	1000–100 km	Telephone system for analog subscriber lines
VLF (Very Low Frequency)	3–30 kHz	100–10 km	Long-range navigation; submarine communications
LF (Low Frequency)	30–300 kHz	10–1 km	Long-range navigation; submarine communication radio beacons
MF (Medium Frequency)	300–3000 kHz	1000–100 m	AM broadcasting; Maritime radio; Direction finding radio
HF (High Frequency)	3–30 MHz	100–10 m	Long-distance aircraft and ship communication; Military communication; Amateur radio
VHF (Very High Frequency)	30–300 MHz	10–1 m	FM broadcasting; Two-way radio; VHF television; Aircraft navigational aids
UHF (Ultra High Frequency)	300–3000 MHz	100–10 cm	UHF television; Cellular mobile telephone; Microwave links; Radar; Personal communications systems (PCS)
SHF (Super High Frequency)	3–30 GHz	10–1 cm	Wireless local loop; Satellite communication; Radar; Terrestrial microwave links
EHF (Extremely High Frequency)	30–300 GHz	10–1 mm	Wireless local loop; specialized laboratory experiments
Infrared Light	300 GHz–300 THz	1 mm–1 nm	Infrared LANs; Consumer electronic applications; Astronomy
Visible Light	400–750 THz	0.75–0.40 nm	Optical fiber communications

Q. 1.2 Explain the basic structure of optical cable..... (4 Marks)

The signal is transmitted as light rays through the fiber optic cable. The light is electromagnetic and its frequency is about 3×10^4 Hz. This is 10^5 times the ultra-high frequency used for transmission of TV signals. So the information carried by light is much more than TV signals. The wavelength of light is small. So the diameter of fiber optic cable is also small. This diameter is comparable to the diameter of human hair.

The basic structure of fiber optic cable is shown in the following figure. The core is the innermost layer. Through the core layer, the light rays travel. The cladding layer covers the core layer. The refractive index of the cladding layer is less than that of the core layer. This is achieved while manufacturing the fiber optic cable. The thickness of a cladding layer is made one or two times the wavelength of light to be guided.



The cladding layer is used to perform the following functions :

- It gives strength to the fiber optic cable.
- This layer acts as a mirror. So the light rays get reflected from it with no power loss.
- When the fiber optic cables are used in a bundle the cladding layer prevents the light rays to get escaped to nearby fiber optic cables.

The last layer is called the jacket or protecting layer. This provides the strength to the fiber optic cable. Also, this layer protects against variations in environmental conditions. Many times a **Kevlar layer** is added to the jacket layer. This gives added strength to the fiber optic cable. Thus in the fiber optic cable, the light rays get reflected from the boundary of the cladding wall and enter into the core layer. So total internal reflections of the light rays occur inside the fiber optic cable.

Q. 1.3 Explain three operating windows in optical communication. (4 Marks)

The light frequency spectrum is divided into three general frequency bands: Infrared, Ultraviolet, and Visible.

Infrared Band:

Infrared is a band of light frequencies that is quite high and cannot be seen by the human eye. Typical useful wavelength range is between 770 nm and 1600 nm. In the infrared spectrum, there are three regions (850 nm, 1300 nm, and 1550 nm) for which silica glass fibers are relatively efficient. Optical fiber systems generally operate in the infrared band.

Visible Band:

This is the band of light frequencies (typically 390–770 nm wavelength range) visible to the human eye. Silica glass fibers are not very good transmitters in the visible spectrum. They attenuate the light waves. So only short optical transmission links are useful.

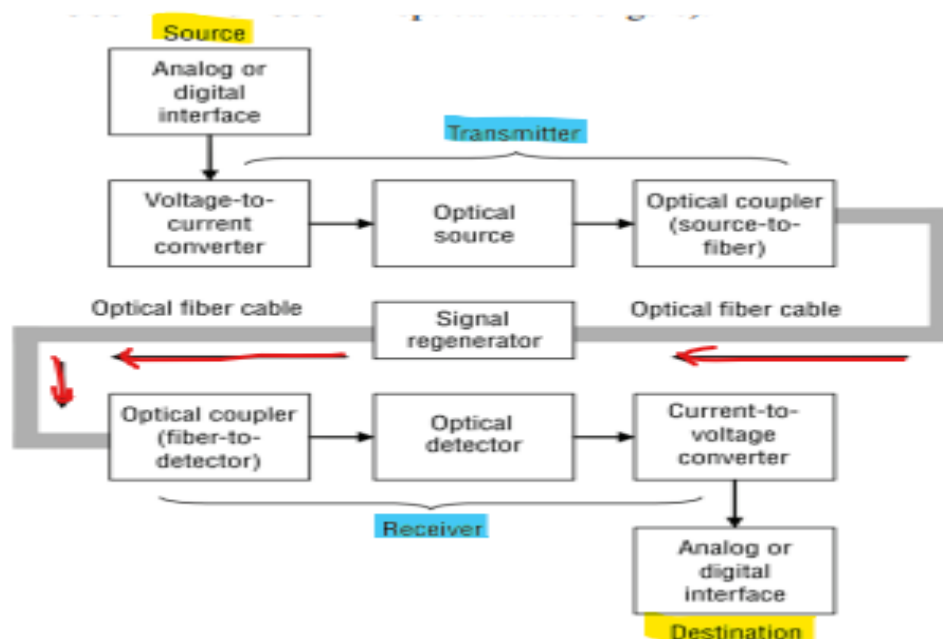
Ultraviolet Band:

The human eye cannot see this frequency band. Typical wavelengths range between 10 nm and 390 nm. The fiber losses in the ultraviolet spectrum are even greater. This band is used in medical applications

Early technology made use of wavelengths from 800 nm to 900 nm. Because for this band of wavelength; optical sources and detectors were easily available. Similarly, the attenuation of data passing through optical fiber was also low. This particular band of wavelength is also called the **first window**. Later, the manufacturers could fabricate optical waveguides in the 1100 nm to 1600 nm region. These waveguides were having very low losses. Here two windows are present. The **second window** is centered around 1300 nm and the **third** around 1500 nm. New fiber materials are used in the 3 μ m to 5 μ m wavelength band. These materials have very low losses. However, it is difficult to manufacture long lengths of fibers.

Q. 1.4 Draw and explain the optical fiber communication system showing all the key elements. Enlist the advantages of optical fibers as a communication media. (MARCH-24,5 Marks)

The following figure shows a simplified lightwave communication system block diagram. It is also known as an optical fiber communication system.



Source: The source of information may be non-electrical or physical, such as voice, image, or video.

Voltage-to-current converter: It is an electrical interface between the information source circuit and the optical (light) source. The amount of light emitted by the light source is generally proportional to the amount of its drive current. Thus, the voltage-to-current converter is necessary to convert an input signal voltage to a current. This current is used to drive the light source.

Optical source: In an optical transmitter, the optical carrier signal generated by an optical source is modulated by an analog or a digital signal. The optical source is a light-emitting diode (LED) or an Injection laser diode (ILD). It generates an electromagnetic wave in the infrared region of the optical spectrum. The input signal modulates the light intensity.

Optical couplers: The function of the source-to-fiber coupler is to collect the light signals from the optical source and send it efficiently to the optical fiber cable. Similarly, the fiber-to-detector coupler is used at the other end of the fiber cable to direct the received light signals onto the photodetector.

Optical fiber cable: It is the guided transmission medium. It is either an ultrapure glass or a plastic cable. The optical fiber consists of a glass or plastic fiber core surrounded by a cladding and then encapsulated in a protective jacket.

Signal regenerators: As the optical signals propagate along the lengths of the optical fiber cable from the source to the destination, they get attenuated (due to absorption, scattering, etc.) and broadened (due to dispersion). As a result, the signals may weaken after traveling a certain distance. Optical regenerators (or optical amplifiers, such as erbium-doped fiber amplifiers) are used at appropriate distances from the transmitter along the length of the fiber cables which help to restore the strength and shape of a transmitted signal.

Optical detector: The optical detector is generally a p-i-n (p-type-intrinsic-n-type) diode, an avalanche photodiode (APD), or a phototransistor. It converts an input optical signal into an equivalent electrical signal.

Current-to-voltage converter: It transforms photodetector current variations to corresponding voltage variations. It produces an output voltage which is proportional to the source information.

Destination (output): Finally, the received information is presented in a form similar to the input information source and suitable for destination devices.

Advantages of Optical Fiber

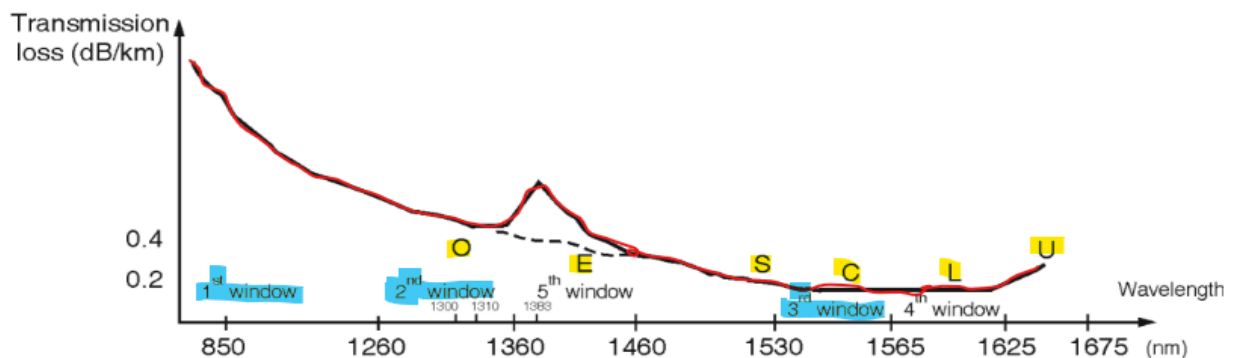
- Larger bandwidth and greater information capacity.
- Lower transmission loss.
- Optical fiber cables provide much higher data integrity and security than metallic cables.
- Since fiber cables are nonconductors of electrical current, they do not radiate electromagnetic energy.
- Optical fibers are made of glass and plastic materials. These fibers are known to be nonconductors of electricity. Therefore, they are immune to crosstalk.
- Optical fiber cables tend to be more resistant to environmental and climatic conditions (including weather variations) than metallic cables.

- Optical fiber cables can also operate over a wider temperature range.
- Optical fiber cables are more reliable than metallic cables and last longer because they exhibit higher tolerance to changes in environmental conditions and are immune to corrosive materials including liquids and gases.
- Easier to install and maintain.

Q. 1.5 Explain the key elements of the optical fiber communication system with a neat diagram and identify the three transmission windows on the fiber attenuation curve..... (FEB-23,6 Marks)

Note: For block diagram and explanation, refer to Q. 1.4

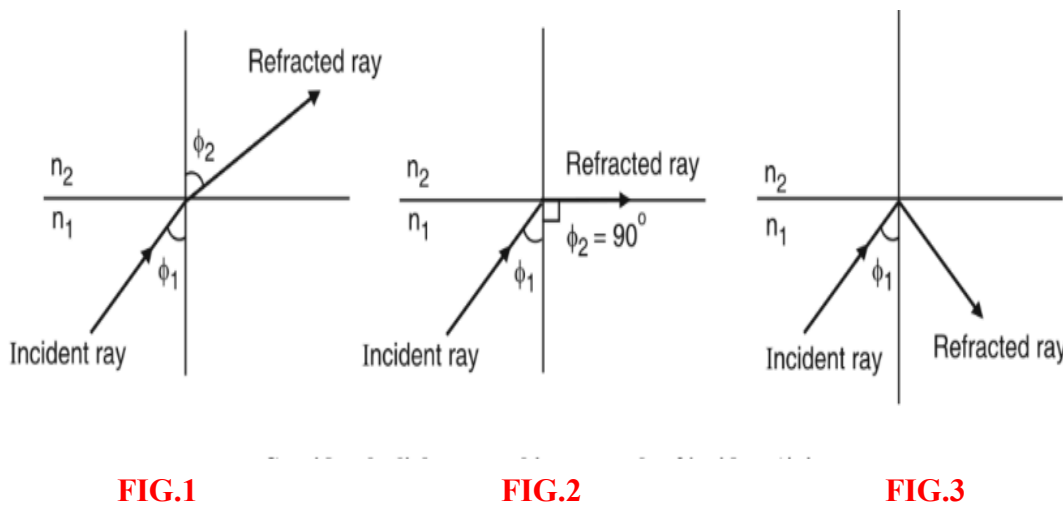
An optical attenuation curve is a plot of transmission loss versus wavelength. It is shown in the following figure. The **first window** is from 800 nm to 900 nm. The **second window** is centered around 1300 nm and the **third** around 1500 nm. The ITU-T organization divided the fiber optical communication system into six bands O, E, S, C, L, and U. Multi-mode optical fiber at 850nm is known as the first window. Single-mode optical fiber at the O band is referred to as the second band. The C band is called the third window, the L is the fourth window, and the E is the fifth.



Q. 1.6 Explain the concept of critical angle..... (4 Marks)

Definition: It is the minimum possible angle of incidence at which if a light ray is incident at the intersection of two different mediums, it gets refracted with an angle of refraction exactly equal to 90° . This phenomenon is usually called the critical angle refraction.

Consider the light ray making an angle of incident ' ϕ_1 '. Fig. 1 shows an incident angle less than critical. Fig. 2 shows incident angle = Critical angle. Figure 3 shows an incidence angle is greater than a critical angle.



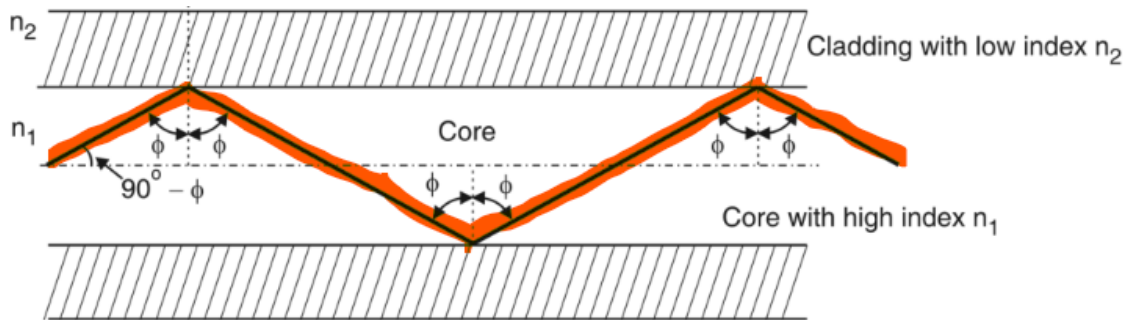
The light ray gets reflected by making an angle of refraction ' ϕ_2 '. Here n_1 and n_2 are the refractive indices of two mediums. The critical angle is denoted by ϕ_c and it is given by

$$\phi_c = \sin^{-1} \left(\frac{n_2}{n_1} \right)$$

Q. 1.7 Using simple ray theory, discuss the mechanism for transmitting light within an optical fiber. With the help of a suitable diagram discuss what is meant by the acceptance angle and numerical aperture of the fiber..... (FEB-23,5 Marks)

The light rays (optical signals) are incident into the cylindrical fiber which is automatically guided (travel) over long distances by an optical phenomenon. It is known as total internal reflection.

If the incidence angle is higher than the critical angle, then the light ray gets reflected rather than causing refraction. This is shown in the following figure.



Inside the fiber optic cable, the incident light ray should cause the total internal reflections. If the refraction of light rays occurs, then the light rays will not enter the fiber optic cable. In this case, the light rays will escape.

Thus, the total internal reflection occurs at the interface between two dielectrics having different refractive indices; when the light is incident on the dielectric of lower index from the dielectric of higher index and the angle of incidence is greater than the critical angle.

Consider a light ray having an angle of incidence ϕ at the interface of core and cladding as shown in Fig. This ray is reflected at the same angle ϕ and it causes the total internal reflection. In this case, the light ray propagates through optical fiber by keeping the shallow angle (less than $90 - \phi_c$). This is called a **meridional ray**. It passes through the axis of the fiber core.

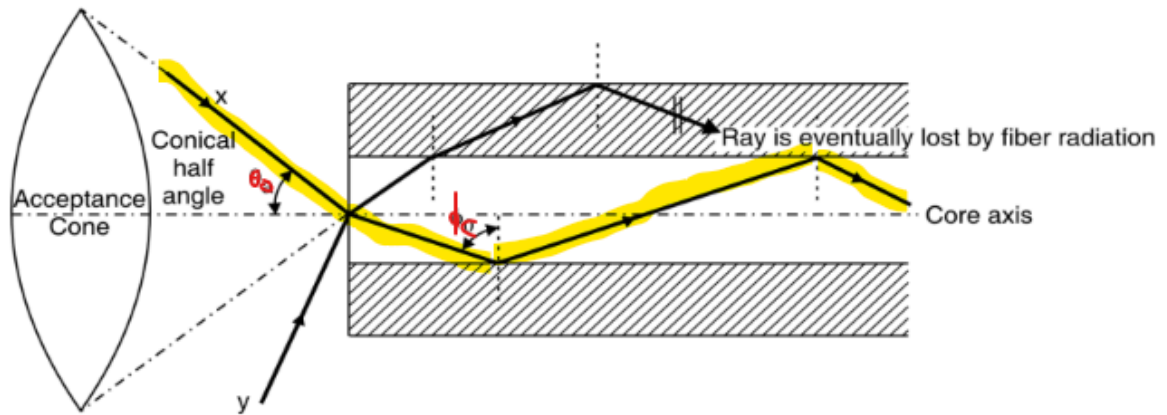
Internal Reflection means when the light beam incident at the core-cladding junction of an optical fiber cable with angles of incidence greater than the critical incidence angle (ϕ_c), it gets reflected into the fiber core with high efficiency.

Acceptance Angle:

The maximum angle at which light enters the fiber and causes internal reflection is called as acceptance angle of an optical fiber. Consider the geometry shown in the following figure.

The light ray 'x' makes an angle θ_a with the core axis. It enters into the fiber cable. It is called a meridional ray. The same ray makes an angle ϕ_c at the core-cladding interface, inside the optical cable. Here ϕ_c is the critical angle, and it is the limiting angle for total internal reflection.

If the angle of incidence is greater than θ_a then the corresponding angle at the core-cladding interface, inside the cable, will be less than that ϕ_c . In this case, the ray will not cause internal reflection.



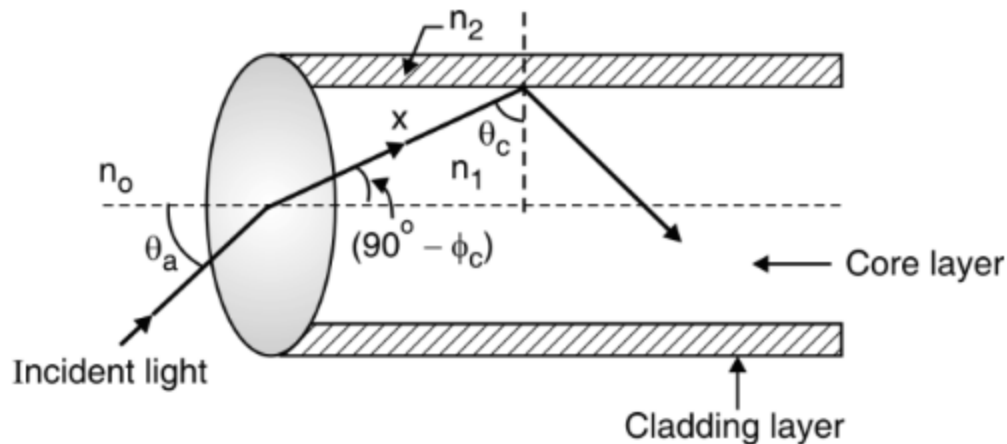
Consider a ray 'y' shown in the figure. It makes an angle greater than θ_a . So the corresponding angle inside the cable will be less than ϕ_c and there is refraction, instead of reflection.

So, this ray enters into the cladding layer and it is eventually lost due to radiation. Thus acceptance angle is the maximum angle of an incoming light ray, with respect to the fiber axis, at which the light may enter into optical cable and gets propagated.

Numerical Aperture:

The numerical aperture (NA) is defined as the sine of the maximum angle that an incident light ray can make with the central axis of the fiber core and can travel through the fiber using the principle of total internal reflection (TIR).

Consider the light ray that enters inside the fiber optic cable by making an angle θ_a with respect to the axis of the fiber optic cable. This light ray strikes the boundary of the core-cladding interface and makes an angle equal to the critical angle. This situation is shown in the following figure.



Here n_0 = Refractive index of air = 1

n_1 = Refractive index of the core

n_2 = Refractive index of cladding

and θ_a = Angle of incidence.

The numerical aperture is given by,

$$NA = \sqrt{n_1^2 - n_2^2}$$

The NA is related to the angle of acceptance as,

$$NA = \sin \theta_a$$

The relative refractive index difference(Δ) between core and cladding is given by,

$$\Delta = \frac{n_1 - n_2}{n_1}$$

The NA in terms of Δ is given by,

$$NA = n_1 \sqrt{2\Delta}$$

For small angles, the solid acceptance angle in air is denoted by ζ . It is related to the NA as,

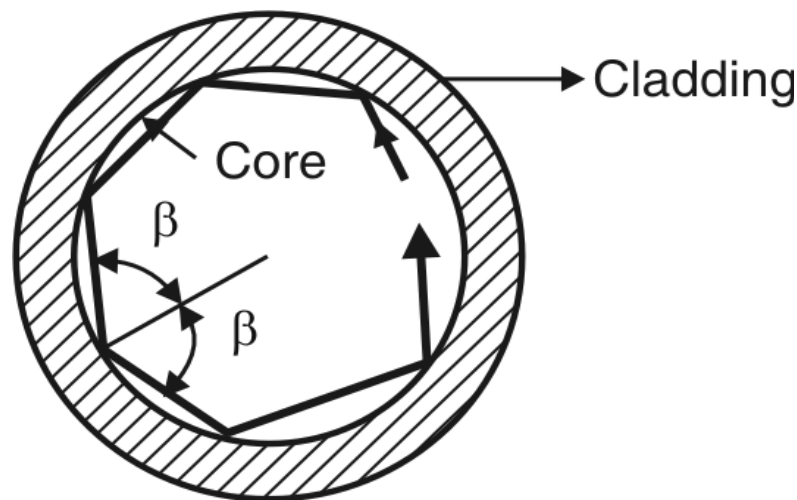
$$\zeta = \pi (NA)^2$$

Q. 1.8 Explain the meridional rays, axial rays, and skew rays in the case of fiber optic cable..... (5 Marks)

If an incoming light ray falls in the cone of acceptance then internal reflection takes place inside the cable. The **meridional rays** enter the core and cause internal reflection by passing through the center of the core.

Some incoming light rays, enter the core and travel along the axis of the core. Such rays are called as **axial rays**.

Some rays follow the helical path inside the core layer as shown in the cross-sectional view of fiber in the following figure. Such rays are called **skew rays**. These rays do not transmit along the axis of the optical cable.



Due to the helical path; tracking of skew rays is difficult. As shown in the figure, β is the angle made by a light ray with the radius. At each reflection, the helical path gives a change in direction of 2β . The point from which, helical rays come out of the fiber depends on the number of reflections inside the cable.

Even if the input light is non-uniform, skew rays produce the smoothing effect. Skew rays change the power losses when they travel through the optical cable. Due to the helical path followed by skew rays, the light acceptance ability of the fiber is changed.

Q. 1.9 A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.50 and a cladding refractive index of 1.47. Determine :(i) The critical angle at the core-cladding interface,(ii) The numerical aperture for fiber, and (iii) The acceptance angle in the air for the fiber..... (5 Marks)

Given: Core refractive index (n_1) = 1.50

Cladding refractive index (n_2) = 1.47

(i) Critical angle:

Formula $\Rightarrow \phi_c = \sin^{-1} \left(\frac{n_2}{n_1} \right)$

$$\therefore \phi_c = \sin^{-1} \left(\frac{1.47}{1.50} \right)$$

$$\therefore \phi_c = 78.52^\circ$$

(ii) Numerical Aperture:

Formula $\Rightarrow \text{NA} = \sqrt{n_1^2 - n_2^2}$

$$\therefore \text{NA} = \sqrt{1.50^2 - 1.47^2}$$

$$\therefore \text{NA} = 0.2984$$

(iii) Acceptance angle:

Formula $\Rightarrow \text{NA} = \sin \theta_a$

$$\therefore \theta_a = \sin^{-1} \text{NA} = \sin^{-1} (0.2984)$$

$$\therefore \theta_a = 17.315^\circ$$

Q. 1.10 A step-index fiber has a relative refractive index difference of 1.3% & a core refractive index of 1.5. The core diameter is 100 μm and the operating wavelength is 850nm. Assuming fiber is kept in air, calculate the numerical aperture of fiber, acceptance angle, and critical angle..... (MARCH-24,6 Marks)

Given: Relative index difference (Δ) = 1.3% = $\frac{1.3}{100} = 0.13$, Core refractive index (n_1) = 1.50

Core diameter = 100 μm $\therefore$ Core radius (a) = 50 $\mu\text{m} = 50 \times 10^{-6} \text{m}$

Wavelength (λ) = 850 nm = $850 \times 10^{-9} \text{m}$

First, we will calculate the value of the refractive index of cladding (n_2).

Formula $\Rightarrow \Delta = \frac{n_1 - n_2}{n_1}$

$$\therefore 0.013 = \frac{n_1 - n_2}{n_1}$$

$$\therefore 0.013 n_1 = n_1 - n_2$$

$$\therefore n_2 = n_1 - 0.013n_1 = n_1(1 - 0.013) = 0.987n_1 = 0.987 \times 1.50$$

$$\therefore n_2 = 1.48$$

Numerical Aperture:

$$NA = \sqrt{n_1^2 - n_2^2}$$

$$\therefore NA = \sqrt{1.50^2 - 1.48^2}$$

$$\therefore NA = 0.244$$

Acceptance angle:

We have the equation,

$$NA = \sin \theta_a$$

$$\therefore \theta_a = \sin^{-1} NA = \sin^{-1} (0.244)$$

$$\therefore \theta_a = 14.12^\circ$$

Critical angle:

$$\phi_c = \sin^{-1} \left(\frac{n_2}{n_1} \right)$$

$$\therefore \phi_c = \sin^{-1} \left(\frac{1.48}{1.50} \right)$$

$$\therefore \phi_c = 80.63^\circ$$

Q. 1.11 A typical relative refractive index difference for an optical fiber designed for long-distance transmission is 1%. Estimate the NA and the solid acceptance angle in the air for the fiber when the core index is 1.46. Further, calculate the critical angle at the core-cladding interface within the fiber. It may be assumed that the concepts of geometric optics hold for the fiber..... (5 Marks)

Given: Relative index difference(Δ) = 1% = $\frac{1}{100} = 0.01$, Core refractive index (n_1) = 1.46

Numerical Aperture:

$$\text{Formula} \Rightarrow NA = n_1 \sqrt{2\Delta}$$

$$\therefore NA = 1.46 \sqrt{2 \times 0.01}$$

$$\therefore NA = 0.2064$$

Solid Acceptance Angle:

Formula $\Rightarrow \zeta = \pi (NA)^2$

$$\therefore \zeta = \pi (0.2046)^2$$

$$\therefore \zeta = 0.133 \text{ rad}$$

Critical angle:

The relative refractive index is given by

Formula $\Rightarrow \Delta = \frac{n_1 - n_2}{n_1}$

$$\therefore \Delta = 1 - \frac{n_2}{n_1}$$

$$\therefore \frac{n_2}{n_1} = 1 - \Delta = 1 - 0.01 = 0.99$$

The critical angle is given by

Formula $\Rightarrow \phi_c = \sin^{-1} \left(\frac{n_2}{n_1} \right)$

$$\therefore \phi_c = \sin^{-1} (0.99)$$

$$\therefore \phi_c = 81.89^\circ$$

Q. 1.12 Explain the cutoff condition of guided modes in optical fiber. Also, define the 'V' number (5 Marks)

The propagation constant is denoted by β . For a mode, to remain as a guided mode in the core layer the condition is,

$$n_2 K < \beta < n_1 K$$

Here n_2 = Refractive index of cladding,

n_1 = Refractive index of the core

and $K = 2\pi/\lambda$, where λ is the wavelength of light

The cut-off condition for such modes is defined as,

$$\beta = n_2 K$$

So when β becomes less than $n_2 K$, there is a power leakage from the core layer to the cladding layer. Thus at this stage, the leaky mode starts.

To summarize, we can write the conditions for different modes as follows :

(i) $n_2 K < n_1 K$; for guided modes

(ii) $n_2 K = \beta$; for cutoff

(iii) $\beta < n_2 K$; for leaky modes.

‘V’ Number: The number of propagating modes in an optical fiber is decided by the ‘V’ number. This value also decides the point at which the mode cutoff condition occurs.

The equation of V in terms of NA is

$$V = \frac{2\pi}{\lambda} a \text{ (NA)}$$

In terms of relative index difference, it is given by,

$$V = \frac{2\pi}{\lambda} a n_1 \sqrt{2\Delta}$$

Q. 1.13 Explain the classification of optical fiber. (6 Marks)

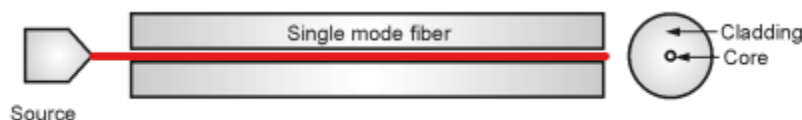
A] Based on operating modes:

According to the operating modes fiber optic cables are classified as :

1) Single-mode fiber optic cables. 2) Multimode fiber optic cables.

1) Single-mode fiber optic cables:

These fiber optic cables carry only one mode. Single-mode fiber has a much smaller core that forces the light to travel in one ray or mode (a single mode) with little light reflection. So the signal travels further as shown in the following figure.

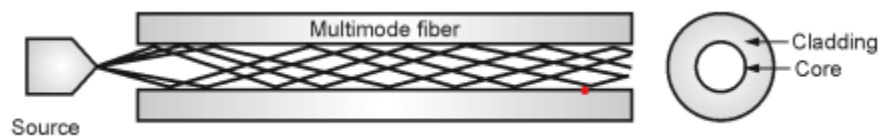


These are designed for a single specific wavelength so that the attenuation losses are minimal. In the case of these fiber optic cables, micro-bending losses are also minimal. Micro-bending losses

are the losses that occur in the fiber optic cable due to lateral displacement of the fiber optic cable.

2) Multimode fiber optic cables:

Light travels through a core in the form of many rays, called modes (multiple modes). These rays are reflected from the cladding surface back into the core as they move through the fiber. This multimode propagation is shown in the following figure.



These fiber optic cables allow different wavelengths to pass through it. Because of this, dispersion losses take place. That means all the wavelengths of light signal don't appear at the output, at the same instant. There is a transit time difference between these signals. These wavelengths are called intermodally dispersed wavelengths.

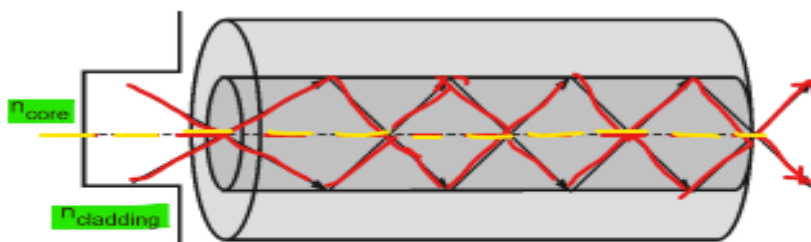
B) Based on Refractive Index Profile:

Depending upon the refractive index profile, there are again two types :

1. Step index fiber optic cables.
2. Graded index fiber optic cables.

1) Step index fiber optic cables:

Step index means a sharp step in the index of refraction between the core and cladding interface as shown in the following figure.

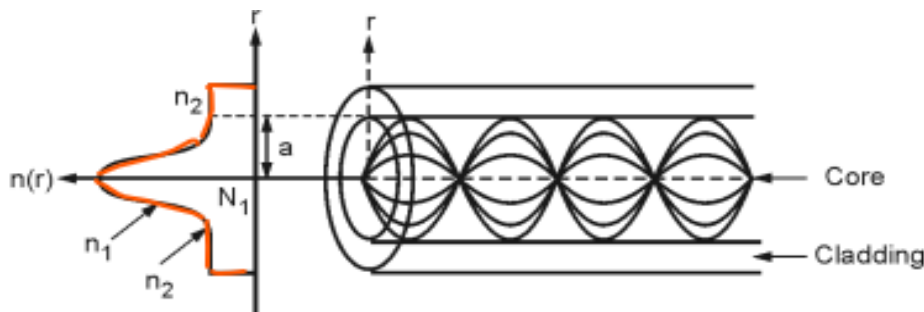


In this case, there is a sudden change in the refractive index while passing from the core to the cladding layer. Because of this abrupt change in the refractive index, the dispersion losses will be more.

2) Graded index fiber optic cables:

The refractive index of the core is non-uniform. This non-uniformity exists because the refractive index is higher at the axis of the core and continuously reduces with the radial movement away from the axis.

In this case, there is a variation in the refractive index while passing from the core layer to the cladding layer. Thus dispersion losses are less compared to step index fibers. The nature of the refractive index of the core is parabolic. This type of fiber optic cable allows greater bandwidth for the signal. The index profile of the step index and graded index fiber is shown in the following figure.



Q. 1.14 Describe with the aid of simple ray diagrams, MMSI and SMSI fiber. Compare advantages and disadvantages of these two types of fibers.. (FEB-23,5 Marks)

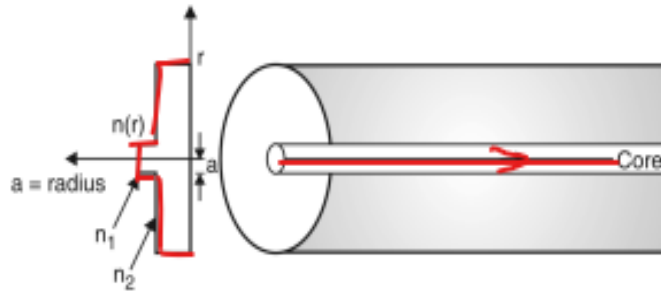
Single Mode Step Index Fiber(SMSI):

A single mode or mono mode step index fiber allows the propagation of only one transverse electromagnetic mode. The V number for step index fiber is 2.405.

The refractive index profile of step index fiber is defined as,

$$n(r) = n_1 \text{ for } r \leq a \text{ And } n(r) = n_2 \text{ for } r \geq a$$

where $n(r)$ represents the refractive index profile of a step-index fiber as a function of its radius (r) , and 'a' denotes the radius of the fiber core. It is shown in the following figure.



Advantages:

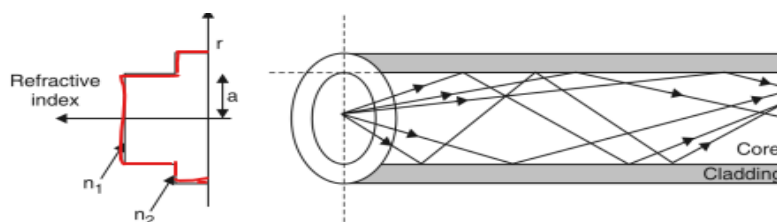
- Since only one mode is propagated; there is no energy loss through leakage due to the higher modes.
- At the output; there is less broadening (dispersion) of a pulse. That means less intermodal dispersion.
- Less attenuation of the signal in the fiber.
- The duration of the output pulse is the same as that of an input pulse.
- The data can be transmitted with a higher bandwidth (above 400 MHz / km).

Disadvantages:

- Since the core diameter is small, the coupling of light rays from a light source to the core layer is difficult.
- The relative refractive index difference between core and cladding is less; so fabrication of such fibers is difficult.

Multimode Step Index Fiber(MMSI):

The 'V' number of multimode fiber is greater than 1.1505, so this fiber supports different modes. The refractive index profile and many modes propagating through multimode step-index fiber are shown in the following figure.



The number of modes propagating through the multimode step index fiber is $M = \frac{V^2}{2}$

Advantages:

- Due to the large core radius, optical power launching in the fiber is easy.
- Connecting two fibers is easy compared to single-mode fibers.
- LED can be easily coupled to the fiber as a light source.
- Compared to single-mode fiber, there is a large refractive index difference between core and cladding layers. So, the fabrication of such fibers is easy.

Disadvantages:

- The different modes follow the different paths. They do not reach the output at the same time. It produces intermodal dispersion.
- The spreading of the output pulse takes place.
- For multimode step index fiber, bandwidth is limited. It is only 6 to 50 MHz/km.
- Due to the large intermodal dispersion, multimode fibers are not suitable for long-distance communication.

Q. 1.15 Compare single-mode and multimode fibers. (MARCH-24,4 Marks)

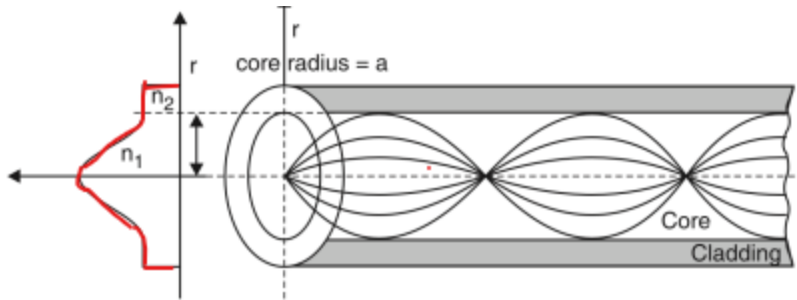
Parameters	Single-mode fiber	Multimode fiber
1) Propagating Mode	Propagation of only one mode takes place	Many modes propagate at the same time.
2) Diameter	Due to the small core diameter; launching of optical power is difficult.	Due to the large core diameter, launching of optical power is easy.
3) Intermodal Dispersion	Due to the single mode, there is no intermodal dispersion.	Many modes travel with different speeds; so intermodal dispersion occurs.

4)Pulse Duration	The duration of the input and output pulse is the same.	The output pulse is spread out compared to the input pulse.
5)Source	LED cannot be used as light source.	LED can be used as a light source
6)Attenuation	At 850 nm, attenuation is about 2 to 5dB/km.	At 850 nm, attenuation is about 2.6 to 50 dB/km.
7)Bandwidth 8)V Number	Higher V number is less than 1.15.	Less V number is greater than 1.15.
9)Distance	Applicable to long-distance communication.	Applicable for short-distance communication.
10)Application	Long-distance telephony, Television broadcasting	LAN networks and video surveillance

Q. 1.16 Sketch and explain the Refractive Index Profile of Graded Index Fiber (GIF) for different values of α (5 Marks)

For GIF, the refractive index of the core layer is not constant. It varies smoothly and continuously over the diameter of the core layer. The refractive index changes proportionally as the distance varies from the core axis to the cladding layer.

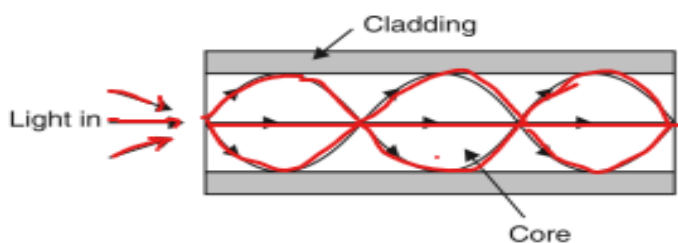
Since the refractive index of the core varies, it is called inhomogeneous core fiber. On the axis of the core, the refractive index is maximum. It is denoted by n_1 . The refractive index starts decreasing with the radial distance as shown in the following figure.



In graded-index multimode fiber, the light travels forward like sinusoidal oscillation. Similar to the step-index multimode fibers, different lights in a graded-index multimode fiber travel along different paths. However, the speed of light propagation in graded-index multimode fibers is different because the speed of guided light varies with the refractive index of the fiber core.

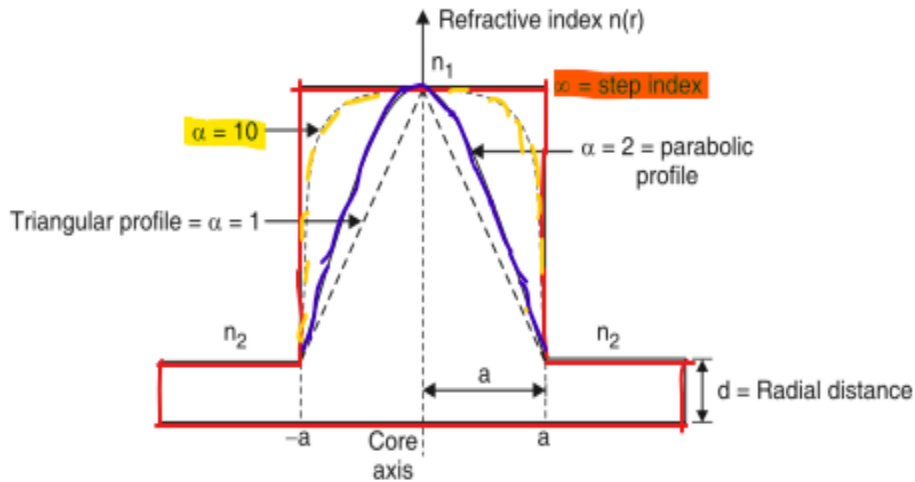
The farther the light goes from the center of the fiber, the faster its speed is. The speed difference compensates for the longer paths followed by the light rays that go farthest from the center of the fiber.

The refractive index decreases from the core axis to the cladding layer. But in the cladding layer; the refractive index is constant. It is denoted by n_2 . Due to the continuously changing refractive index, the propagation of light rays is not along a straight line path. Consider the propagation through multimode-graded index fiber as shown in the following figure.



For the curved path, the velocity of propagation changes for a change in refractive index. At the center of the core, the refractive index is maximum. So, the axial ray passing along the center of the core has the slowest speed. Due to the difference in speeds of different paths; even though the distance traveled by each path is different, all paths reach the output at the same time.

Let ' α ' denotes the profile parameter, which gives the characteristic refractive profile of the core layer. The value of ' α ' decides the shape of the refractive index profile curve. The refractive index profile for different values of ' α ' is shown in the following figure.



Q. 1.17 If a multimode step index fiber has a core refractive index of 1.5, a cladding refractive index of 1.38, and a core radius of $25\ \mu\text{m}$ operates at a wavelength of $1300\ \text{nm}$. Calculate :

- (i) Numerical aperture.
- (ii) Normalized frequency.
- (iii) Solid acceptance angle.
- (iv) Total no. of modes entering the fiber..... (6 Marks)

Given : $n_1 = 1.5$, $n_2 = 1.38$, $a = 25\ \mu\text{m} = 25 \times 10^{-6}\ \text{m}$, $\lambda = 1300\ \text{nm} = 1300 \times 10^{-9}\ \text{m}$

(i) Numerical Aperture (NA):

Formula $\Rightarrow \text{NA} = \sqrt{n_1^2 - n_2^2}$

$$\therefore \text{NA} = \sqrt{1.5^2 - 1.38^2}$$

$$\therefore \text{NA} = 0.5878$$

ii) Normalized Frequency(V):

Formula $\Rightarrow V = \frac{2\pi}{\lambda} a (\text{NA})$

$$\therefore V = \frac{2\pi}{1300 \times 10^{-9}} \times 25 \times 10^{-6} \times 0.5878$$

$$\therefore V = 71.024$$

(iii) Acceptance angle:

Formula $\Rightarrow \theta_a = \sin^{-1} NA = \sin^{-1} (0.5878)$

$\therefore \theta_a = 36^\circ$

(iv) Solid acceptance angle (ζ):

$$\zeta = \pi (NA)^2$$

$$\therefore \zeta = \pi (0.5878)^2$$

$\therefore \zeta = 1.0854 \text{ rad}$

(v) Total number of modes (M_g) :

Formula $\Rightarrow M_g = \frac{V^2}{2}$

$$\therefore M_g = \frac{71.024^2}{2}$$

$\therefore M_g = 2522.2 \approx 2523 \text{ modes}$

Q. 1.18 A manufacturer wishes to make a silica-core, step-index fiber with $V = 75$ and a numerical aperture $NA = 0.30$ to be used at 820 nm . If $n_1 = 1.458$.
What should the core size and cladding refractive index be?..... (FEB-23,5 Marks)

Given : $V = 75$, $NA = 0.30$, $n_1 = 1.458$, $\lambda = 820 \text{ nm} = 820 \times 10^{-9} \text{ m}$

Calculation of cladding refractive index (n_2):

The formula of NA is

Formula $\Rightarrow NA = \sqrt{n_1^2 - n_2^2}$

Take a square of both sides,

$$(NA)^2 = (n_1^2 - n_2^2)$$

$$\therefore n_2^2 = n_1^2 - (NA)^2$$

Take a square root of both sides,

$$\therefore n_2 = \sqrt{n_1^2 - (NA)^2}$$

$$\therefore n_2 = \sqrt{1.458^2 - (0.30)^2}$$

$$\therefore n_2 = 1.4268$$

Calculation of core size (a):

The equation of the 'V' number is

$$\text{Formula} \Rightarrow V = \frac{2\pi}{\lambda} a (NA)$$

$$\therefore a = \frac{V\lambda}{2\pi (NA)}$$

$$\therefore a = \frac{75 \times 820 \times 10^{-9}}{2\pi \times 0.30}$$

$$\therefore a = 32.62 \times 10^{-6} \text{ m}$$

$$\therefore a = 32.62 \mu\text{m}$$

Q. 1.19 Calculate the number of modes at 1.3 μm wavelengths in GIF having index with parabolic refractive index profile, core radius 25 μm , core refractive index 1.48, and cladding refractive index 1.46..... (5 Marks)

Given, $\lambda = 1.3 \mu\text{m} = 1.3 \times 10^{-6} \text{ m}$, $\alpha = 2$, $a = 25 \mu\text{m} = 25 \times 10^{-6} \text{ m}$, $n_1 = 1.48$, $n_2 = 1.46$

Total modes in GIF having parabolic refractive index,

$$\text{Formula} \Rightarrow M_g = \frac{V^2}{4}$$

The equation of the 'V' number is,

$$V = \frac{2\pi}{\lambda} a (NA) \text{ and } NA = \sqrt{n_1^2 - n_2^2}$$

$$\therefore V = \frac{2\pi}{\lambda} a \sqrt{n_1^2 - n_2^2}$$

$$\therefore V = \frac{2\pi}{1.3 \times 10^{-6}} \times 25 \times 10^{-6} \times \sqrt{1.48^2 - 1.46^2}$$

$$\therefore V = 29.2998$$

$$\therefore M_g = \frac{29.2998^2}{4}$$

$$\therefore M_g = 214.62 \approx 215 \text{ modes}$$

Q. 1.20 Classify and explain various loss mechanisms in optical fibers..... (FEB-23,5 Marks)

OR

Describe the following losses in optical fiber.

i) Absorption

ii) Attenuation

iii) Scattering (MARCH-24,5 Marks)

Losses in optical fibers result from attenuation in the material itself and from scattering. The losses are classified as

1. Absorption loss
2. Scattering (Linear and non-linear) losses
3. Attenuation losses
4. Bending losses
5. Coupling losses

1) Absorption loss:

During the optical cable fabrication, some of the transmitted light is dissipated as heat. It is called material absorption loss. Absorption loss is directly related to the actual composition of the material and the manufacturing process used. The major factors responsible for material absorption loss is as follows :

- (i) Intrinsic absorption due to basic atoms of fiber material.
- (ii) Extrinsic absorption due to impurity atoms.
- (iii) Absorption due to atomic defects in the glass material.

2) Scattering (Linear and non-linear) losses:

Due to the non-uniformities in fiber optic cable, a straight line path of light rays gets deviated. It is referred to as scattering. Scattering loss occurs due to the transfer of the optical power available within one propagation mode into a different propagation mode. The scattering may happen when a light beam propagating within the fiber strikes an imperfection in the core material and then changes its direction. This may result in a power loss.

These losses are mainly caused by the interaction of light with density fluctuations within a fiber. These density fluctuations occur when the fiber is produced. The glass is composed of a

randomly connected network of molecules, which is made up of several oxides and it increases the compositional fluctuations.

In the case of multimode fibers, there is a higher doping concentration and greater compositional fluctuations. Thus scattering losses are more.

3) Attenuation losses:

Attenuation represents a reduction in the amplitude of a signal. In the case of optical cable, it represents the reduction in the intensity of light rays propagating through it. It is mainly caused due to the scattering and absorption of light rays.

Attenuation is expressed in dB. Let an input power launched in the fiber be denoted by P_{IN} and the output power by P_{OUT} then the attenuation for the length 'L' is given by,

$$\alpha = \frac{10}{L} \log \left(\frac{P_{IN}}{P_{OUT}} \right) \text{ dB}$$

4) Bending losses:

Bending an optical fiber too sharply can increase fiber losses by causing some of the light to meet the intersection with the fiber core and the cladding at an angle less than the critical angle. Thus preventing the desired phenomenon of total internal reflection.

If there is an abrupt change in the radius of curvature of the fiber, then radiation loss occurs from the fiber. If there is a sharp bend of the fiber, there is a probability of mechanical failure of the optical cable. Usually, the higher-order modes are not tightly bound to the core layer. So due to the sharp bends, the radiation losses of such modes occur.

5) Coupling losses:

In optical fiber cables, coupling losses can occur at any of the following three types of optical junctions:

- connections between a light source and the fiber at the transmitter end (source-to-fiber power launching)
- fiber-to-fiber interconnections (to extend the length of the optical fibers)
- connections between the fiber and the photodetector at the receiver end

Coupling the fiber to sources and detectors creates losses, especially when it involves mismatches in numerical aperture or the size of optical fibers. In an optical fiber communication system, the losses in splices and connectors can easily be more than in the fiber cable itself.

Coupling losses occur due to the imperfect physical connections in the optical fiber communication system. Junction losses are most often caused by one of the following alignment problems:

1. Lateral or axial misalignment.
2. Gap misalignment
3. Angular misalignment
4. Imperfect surface finishes

Q. 1.21 The mean optical power launched into an optical fiber link is 1.5 mW and the fiber has an attenuation of 0.5dB/Km. Determine the maximum possible link length without repeaters when the minimum mean optical power level required at the detector is 2μW..... (FEB-23,4 Marks)

Given: $P_{IN} = 1.5 \text{ mW} = 1.5 \times 10^{-3} \text{ W}$, $P_{OUT} = 2 \text{ μW} = 2 \times 10^{-6} \text{ W}$, $\alpha = 0.5 \text{ dB/Km}$

The attenuation in an optical cable is given by,

Formula $\Rightarrow \alpha = \frac{10}{L} \log \left(\frac{P_{IN}}{P_{OUT}} \right)$

$$\therefore L = \frac{10}{\alpha} \log \left(\frac{P_{IN}}{P_{OUT}} \right)$$

$$\therefore L = \frac{10}{0.5} \log \left(\frac{1.5 \times 10^{-3}}{2 \times 10^{-6}} \right)$$

$$\therefore L = 57.50 \text{ Km}$$

Q. 1.21 Draw and explain fusion splicing..... (5 Marks)

The splices are used to connect the two fiber optic cables permanently. The basic requirements of splices are :

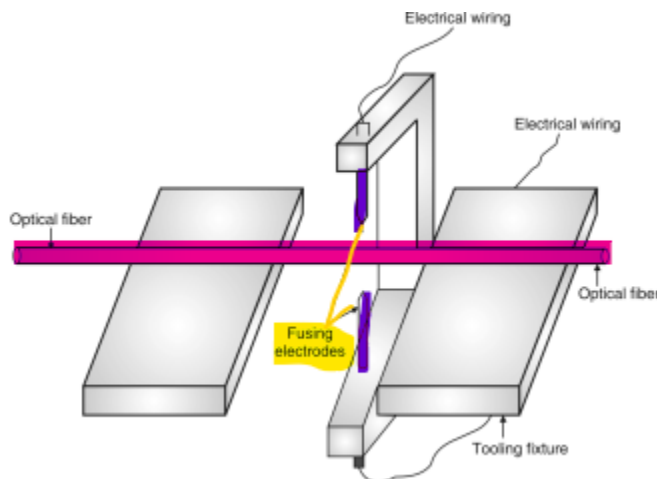
- (i) Splices should cause the minimum power loss.
- (ii) They should be easy to install.
- (iii) They should cause a low attenuation.
- (iv) They should be strong and lightweight.

There are two techniques for splicing optical fibers depending on the insertion loss, cost, and performance characteristics. This splicing technique is:

- (i) Fusion splicing
- (ii) Mechanical splicing.

Fusion Splicing:

Fusion Splicing makes a permanent connection between two optical fiber cables. It gives a longer life with less attenuation. The two cores of fiber cables are joined or fused electrically or thermally. That means an electric device or an electrical arc is used to fuse the two fiber optic cables and produce the connection between them. The following figure shows the fusion splicing of optical fiber technology.



The two fiber cables are aligned together by using a device called a fusion splicer. These cables can be fused or joined together to form a connection with the help of an electric arc. The heat produced by the electric arc can give a transparent and continuous non-reflective connection between two optical fiber cables with less attention, and insertion losses.

Advantages:

1. It provides the lowest attenuation.
2. It provides high-quality joints.
3. It has a small size of splice.
4. It is very compact
5. It has the lowest insertion loss
6. It has the lowest back reflection (optical return loss ORL)

7. It has the highest mechanical strength
8. It is permanent
9. It can withstand extremely high-temperature changes
10. It prevents dust and other contaminants from entering the optical path.

Disadvantages:

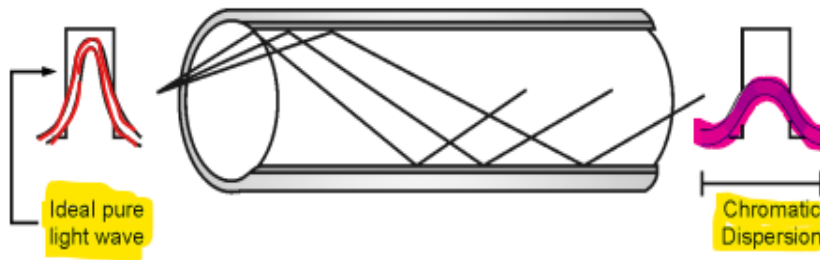
1. The heat is used for splicing. It makes the fiber ends weak.
2. After splicing, the tensile strength becomes low.
3. The cost of fusion splicing is very high due to the expensive fusion splicing device.
4. It requires a constant power supply and some special tools.
5. There are some situations where fusion splicing is not practical or we can say it is impossible, so as an alternative, we have to use mechanical splicing.
6. It is a time-consuming technique and can't be used for temporary connections.
7. Fusion splicers need periodic maintenance which involves regular cleaning, electrode alignment, and occasional replacement.
8. It is mainly used with single-mode fiber, unlike mechanical splicing which works for both single-mode and multi-mode fiber.

Q. 1.22 Explain intramodal dispersion, intermodal dispersion, and overall fiber dispersion in optical fibers..... (MARCH-24,4 Marks)

Definition: Dispersion means spreading out of an optical pulse of light energy in time as it propagates down a fiber. It is also called pulse spreading. Dispersion occurs because of the difference in the propagation time taken by the light rays that traverse different propagation paths within the fiber.

Chromatic or Intramodal Dispersion:

It results from the finite spectral linewidth of the optical source. Propagation delay differences between the different spectral components of the transmitted light signal cause broadening of each transmitted mode. It is collectively known as intramodal dispersion. It is caused by the dispersive properties of the fiber material (material dispersion) and guidance effects within the fiber structure (waveguide dispersion). The following figure shows intramodal dispersion.

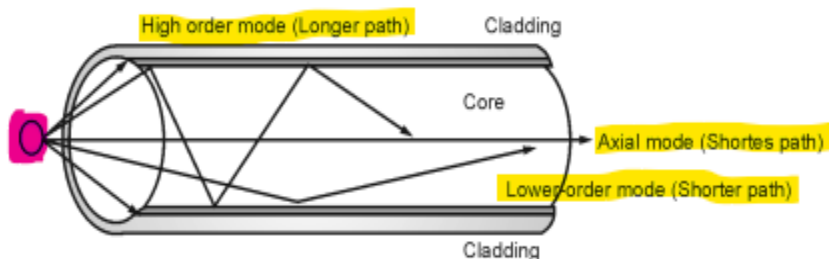


Intermodal Dispersion or Mode Dispersion:

Modal dispersion is a distortion mechanism occurring in multimode fibers and other waveguides, in which the signal is spread in time. This is because of different propagation velocities for all modes.

The light rays entering the fiber at different angles of incidence will go through different paths or modes. Some of these light rays will travel straight through the center of the fiber (axial mode) while others will repeatedly bounce off the cladding-core boundary to follow a zigzag path along the optical cable.

Due to a bounce-off, modal dispersion (or intermodal dispersion) takes place. The high-order modes (light entering at sharp angles) have more modal dispersion than low-order modes (light entering at smaller angles). It is shown in the following figure.



Overall Dispersion:

The overall dispersion in the fiber is the summation of intermodal and intramodal dispersion. The total RMS pulse broadening (σ_T) is given by,

$$\sigma_T = \sqrt{\sigma_c^2 + \sigma_n^2}$$

Here, σ_c = Intramodal broadening and σ_n is Intermodal broadening.

Q. 1.23 A multimode-graded index fiber exhibits a total pulse broadening of 0.1 μ sec over a distance of 15km. Calculate

i) Maximum possible bandwidth on the link

ii) The pulse dispersion per unit length

iii) The bandwidth-length product for the fiber..... (MARCH-24,6 Marks)

Given: Pulse broadening (T)= 0.1 μ sec = 0.1×10^{-6} sec, Length (L)=15 km

(i) The maximum possible bandwidth assuming no intersymbol interference:

Formula $\Rightarrow B_{opt} = \frac{1}{2T}$

$$\therefore B_{opt} = \frac{1}{2 \times 0.1 \times 10^{-6}}$$

$$\therefore B_{opt} = 5 \times 10^6 \text{ Hz}$$

$$\therefore B_{opt} = 5 \text{ MHz}$$

(ii) Dispersion per unit length:

Formula $\Rightarrow$ Dispersion per unit length = $\frac{T}{L}$

$$\therefore \text{Dispersion per unit length} = \frac{0.1 \times 10^{-6}}{15} = 6.66 \times 10^{-9} \text{ s / km}$$

$$\therefore \text{Dispersion per unit length} = 6.66 \text{ ns/km}$$

(iii) The bandwidth length product:

Formula $\Rightarrow$ bandwidth length product = $B_{opt} \times L$

$$\therefore \text{Bandwidth length product} = 5 \times 10^6 \times 15$$

$$\therefore \text{Bandwidth length product} = 75 \text{ MHz.km}$$

Unit 2: OPTICAL SOURCES

Q. 2.1 State the semiconductor materials used for various optical sources

(FEB-23,4 Marks)

The different semiconductor materials used for various optical sources such as LED and Laser Diode are as follows:

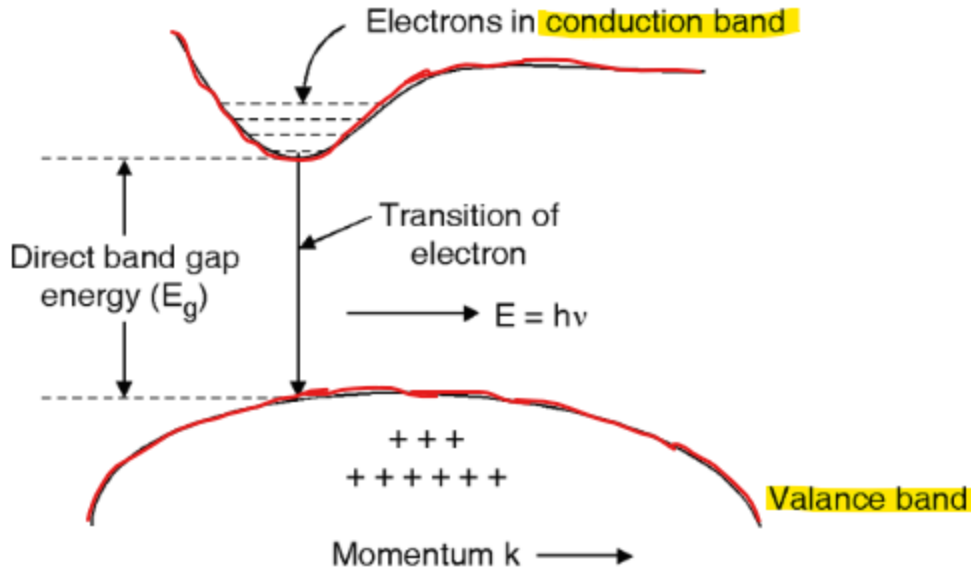
- 1) GaAs (Gallium Arsenide):** Used for infrared LED and Laser diodes. The wavelength range is 850 nm to 940 nm.
- 2) GaP (Gallium Phosphide):** Generally used for the yellow-green region of the visible spectrum. The wavelength range is 550 nm to 570 nm.
- 3) InGaN (Indium Gallium Nitride):** Used for blue, green, and white LEDs. The wavelength range is 400 nm to 550 nm.
- 4) AlGaInP (Aluminum Gallium Indium Phosphide):** Used for red and yellow LEDs. The wavelength range is 590 nm to 660 nm.
- 5) InP (Indium Phosphide):** It is a common semiconductor material used for long-wavelength laser diodes. The wavelength range is 1300 nm to 1550 nm.
- 6) GaInAsP (Gallium Indium Arsenide Phosphide):** It is a common semiconducting material system used for various laser diodes. The wavelength range is 1200 nm to 1600 nm.

Q. 2.2 Distinguish direct and Indirect band-gap semiconductor materials. Also, state the major requirements of good optical sources..... (MARCH-24,6 Marks)

Whenever there is a transition of electrons between a conduction band and a valance band, energy and momentum are conserved. Depending on the shape of the bandgap as a function of momentum, semiconductors are classified as: direct bandgap and indirect bandgap materials.

Direct Bandgap Materials:

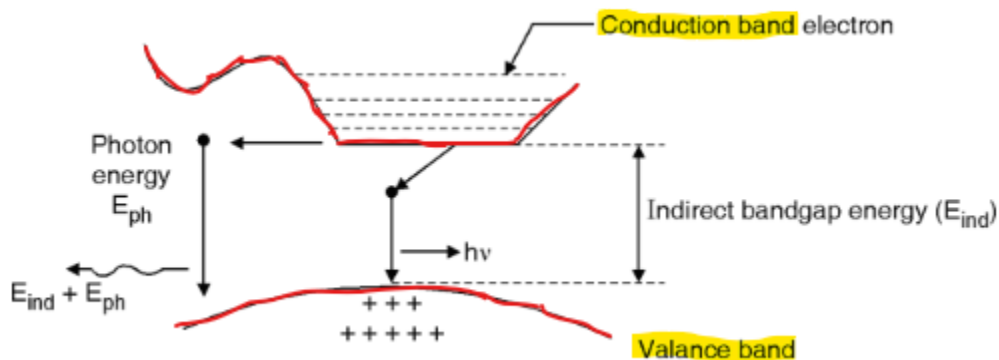
In a direct band gap semiconductor, the top of the valence band and the bottom of the conduction band occur at the same value of momentum. An electron from the conduction band falls and recombines with a hole in the valance band. This falling electron emits energy, which means it emits a photon. This phenomenon is shown in the following figure.



Here electrons and holes have the same momentum value. There is a direct recombination of electrons and holes. Such materials are called direct bandgap materials and have higher internal quantum energy. These semiconducting materials are used for the fabrication of LEDs and laser diodes.

Indirect Bandgap Materials:

Here the conduction band minimum and valence band maximum energy levels occur at different values of momentum. An electron does not directly fall from the conduction band to the valence band, but it falls at an intermediate level before reaching the valence band. The concept of indirect band material is shown in the following figure.



This combination process is relatively slow. Thus such materials have a longer lifetime for charge carriers. By adding impurities, electron-hole recombination can be increased and the carrier lifetime can be effectively reduced.

Major Requirements of Good Optical Sources:

- The emission wavelength should be compatible with the loss spectrum of glass fibers: 820nm, 1300nm & 1550nm.
- The source should be capable of modulation at rates over 1GHz for high data rate transmission.
- The Spectral width of the source should be narrow to minimize the bandwidth limiting pulse dispersion in the fibers.
- The radiance of the source should be as high as possible for effective coupling into the low-loss fiber with a small NA(0.2). This means that the beam spread of the source must be minimized.
- The source must be highly reliable.
- The source should be reasonably low-cost.

Q. 2.3 Calculate the emission wavelength in nm of the light emitted using a semiconductor material with an energy band-gap of 3eV..... (MARCH-24,3 Marks)

Given: Bandgap energy (E_g) = 3eV (electronvolt)

1 eV is equal to 1.602×10^{-19} Joules.

$$\therefore 3eV = 3 \times 1.602 \times 10^{-19} = 4.806 \times 10^{-19} \text{ J}$$

The energy of a photon (E) and wavelength (λ) are related by, $E = \frac{hc}{\lambda}$

Here 'h' is Planck's constant = 6.626×10^{-34}

'C' = speed of light = 3×10^8 m/s

$$\therefore \lambda = \frac{hc}{E}$$

$$\therefore \lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{4.806 \times 10^{-19}}$$

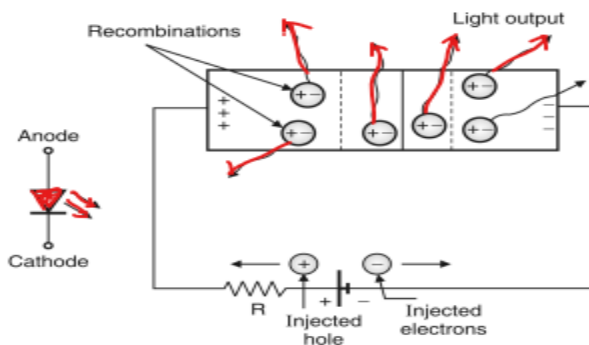
$$\therefore \lambda = 4.136 \times 10^{-7} \text{ m}$$

$$\therefore \lambda = 413.6 \text{ nm}$$

Q. 2.4 Describe with the help of a suitable diagram the mechanism giving the emission of light from an LED. State useful properties of LED to be used in optical fiber communication..... (FEB-23,5 Marks)

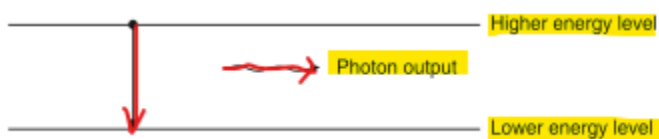
LEDs work on the principle of Electroluminescence. On passing a current through the diode, minority, and majority charge carriers recombine at the junction. On recombination, energy is released in the form of photons.

As the forward voltage increases, the intensity of the light increases and reaches a maximum. Here in the case of LEDs, the battery connection provides a higher-level electron supply. The symbol and the forward-biased LED connection are shown in the following figure.



The lighting emitting diode is a p-n junction diode. It is a specially “doped diode” made up of a special type of semiconductor with a suitable lens. The length of the anode terminal is longer than that of the cathode terminal.

As shown in the figure, the diode is made forward-biased. Here the battery supplies the holes into the ‘p’ region and electrons into the ‘n’ region. These holes and electrons supplied by the battery are called injected charge carriers. Now injected electrons, near the potential barrier get recombined with the holes. That means the electrons are falling from higher energy level to lower energy level as shown in the following figure.



When this transition occurs then the energy is radiated out. When the diode is forward-biased, the electrons & holes move fast across the junction and are combined constantly. Soon after the electrons move from the n-type to the p-type silicon, they combine with the holes, and then it

disappears. Hence it makes the complete atom more stable and gives a little burst of energy. It is in the form of a tiny packet or photon of light.

Useful Properties of LEDs for Optical Fiber Communication:

The following properties of LEDs make them suitable for use in optical fiber communication systems.

- LEDs are compact and can be easily coupled to the core of optical fiber.
- The coupling efficiency is high
- LEDs are less expensive.
- LEDs are highly efficient in converting electrical energy into light.
- LEDs have lower power consumption compared to other light sources.
- The light output of LEDs can be easily modulated.
- LEDs have a wider range of wavelengths.

Q. 2.5 Explain the working and characteristics of LEDs with diagrams. State its advantages and disadvantages..... (MARCH-24,4 Marks)

For the working and characteristics of LED refer to Q. 2.4

Advantages of LED:

1. They are small in size.
2. Light in weight.
3. They are mechanically rugged.
4. They have lower operating temperatures.
5. No complex driver circuitry is required.
6. Switch - on time is very small.
7. They have higher speed.
8. Available in different colors.

9. They have a longer life compared to lamps.
10. Linearity is more.
11. Compatible with integrated circuits.
12. Cost is low.

Disadvantages of LED:

1. Output power is affected by the temperature variation.
2. Quantum efficiency is low.
3. Gets damaged because of overvoltage and overcurrent.
4. LED light causes Blue pollution.
5. LED fixtures are voltage Sensitive.
6. LED fixtures do not allow spherical light Distribution.

Q. 2.6 Explain external quantum efficiency and the external power generated in the LED with necessary equations..... (MARCH-24,6 Marks)

Quantum Efficiency:

The quantum efficiency of an LED is defined as the ratio of the number of photons emitted to the number of electrons injected inside. It is denoted by η .

$$\therefore \eta = \frac{\text{Number of photons emitted out}}{\text{Number of electrons injected in}}$$

The internal quantum efficiency of LED (η_{int}) is given by,

$$\eta_{int} = \frac{\tau}{\tau_r}$$

Here τ = Total recombination lifetime and

$$\tau = \frac{\tau_r \times \tau_{nr}}{\tau_r + \tau_{nr}}$$

Here τ_r is radiative recombination rise time and τ_{nr} is non-radiative recombination rise time. The external quantum efficiency is given by,

$$\eta_{ext} = \frac{1}{n(n+1)^2}$$

Here n = Refractive index of light source material

Most LEDs have a quantum efficiency of less than 1. The efficiency gets reduced because of the following factors :

- (a) Nature of material used.
- (b) Change in refractive index.
- (c) Angle made by emitted photons.

Materials used for the manufacturing of LED may consist of defects or flaws. So the emitted photons can get trapped in these flaws and won't come out. This reduces the total number of output photons and hence the quantum efficiency. The light rays (photons) from the LED enter the air medium. The refractive index of air is 1.

The bending of light rays occurs because of a change in refractive index. This again reduces the quantum efficiency. Some of the light rays coming out from LED make an angle of incidence greater than the critical angle. These light rays cause internal reflections that mean they will not come out from the LED structure.

External Power Generated in LED:

It is the power radiated in the form of light from the LED. This power depends on electrical input power and external quantum efficiency (η_{ext}). An electrical power input to the LED is,

$$P_{in} = V \times I$$

Here V is the applied voltage across the LED and I is the current flowing through the LED.

The output power from the LED is given by,

$$P_{out} = P_{in} \times \eta_{ext}$$

Q. 2.7 Explain radiative and non-radiative recombination lifetimes of LED.....(3 Marks)

When an electron falls from a higher to a lower energy level; energy is radiated. If it gives a photon then it is called **radiative recombination**. If a heat is generated then it is called **non-radiative recombination**.

The average time spent by an electron in the excited state before radiative recombination is called radiative lifetime. The average time spent by an electron in the excited state before non-radiative recombination is called non-radiative lifetime.

The following factors affect lifetimes:

- The band gap energy of a material and defects in the material affect the lifetimes.
- Higher temperature increases non-radiative recombination rates. It reduces the efficiency of LED
- High current densities increase non-radiative recombination rates. It reduces the efficiency of LED.

Q. 2.8 The radiative and non-radiative lifetimes of the minority carriers in the active region of a double heterojunction LED are 60 ns and 100 ns respectively. Determine the total carrier recombination lifetime and the power internally generated within the device when the peak emission wavelength is 0.87μm at a drive current of 40 mA.(FEB-23,6 Marks)

Given: Radiative recombination lifetime (τ_r) = 60 ns = 60×10^{-9} sec,

non-radiative recombination lifetime (τ_{nr}) = 100 ns = 100×10^{-9} sec,

Wavelength (λ) = 0.87 μm = 0.87×10^{-6} m,

Drive current = $i = 40$ mA = 40×10^{-3} A

Total recombination lifetime:

Formula $\Rightarrow \tau = \frac{\tau_r \times \tau_{nr}}{\tau_r + \tau_{nr}}$

$$\therefore \tau = \frac{60 \times 10^{-9} \times 100 \times 10^{-9}}{60 \times 10^{-9} + 100 \times 10^{-9}}$$

$$\therefore \tau = 37.5 \times 10^{-9} \text{ sec.}$$

$$\therefore \tau = 37.5 \text{ nsec}$$

Internally generated power:

Internal quantum efficiency is,

$$\text{Formula} \Rightarrow \frac{\tau}{\tau_r}$$

$$\therefore \eta_{int} = \frac{37.5 \times 10^{-9}}{60 \times 10^{-9}}$$

$$\therefore \eta_{int} = 0.625 = 62.5 \%$$

Internal power is given by,

$$\text{Formula} \Rightarrow P_{int} = \eta_{int} \frac{ihC}{q\lambda}$$

Here h = Plank's constant = 6.625×10^{-34}

C = speed of light = 3×10^8 m/sec

q = Charge of electron = 1.602×10^{-19}

$$\therefore P_{int} = 0.625 \times \frac{40 \times 10^{-3} \times 6.625 \times 10^{-34} \times 3 \times 10^8}{1.602 \times 10^{-19} \times 0.87 \times 10^{-6}}$$

$$\therefore P_{int} = 0.0356 \text{ W} = 35.65 \text{ mW}$$

Q. 2.9 A double heterojunction at a peak wavelength of InGAASP LED emitting at a peak wavelength of 1310 nm has radiative and non-radiative recombination times of 30 and 100 ns respectively. The drive current is 40 mA Determine

i) Bulk recombination lifetime

ii) Internal quantum efficiency

iii) Internal power level..... (MARCH-24,6 Marks)

Given: Radiative recombination lifetime (τ_r) = 30 ns = 30×10^{-9} sec,

non-radiative recombination lifetime (τ_{nr}) = 100 ns = 100×10^{-9} sec,

Wavelength (λ) = 1310 nm = 1310×10^{-9} m,

Drive current = $i = 40$ mA = 40×10^{-3} A

Total recombination lifetime:

$$\tau = \frac{\tau_r \times \tau_{nr}}{\tau_r + \tau_{nr}}$$

$$\therefore \tau = \frac{30 \times 10^{-9} \times 100 \times 10^{-9}}{30 \times 10^{-9} + 100 \times 10^{-9}}$$

$$\therefore \tau = 23.076 \times 10^{-9} \text{ sec.}$$

$$\therefore \tau = 23.076 \text{ nsec}$$

Internal quantum efficiency:

$$\eta_{int} = \frac{\tau}{\tau_r}$$

$$\therefore \eta_{int} = \frac{23.076 \times 10^{-9}}{30 \times 10^{-9}}$$

$$\therefore \eta_{int} = 0.7692 = 76.92 \%$$

Internally generated power:

$$\text{Formula} \Rightarrow P_{int} = \eta_{int} \frac{ihC}{q\lambda}$$

Here h = Planck's constant = 6.625×10^{-34}

C = speed of light = 3×10^8 m/sec

q = Charge of electron = 1.602×10^{-19}

$$\therefore P_{int} = 0.7692 \times \frac{40 \times 10^{-3} \times 6.625 \times 10^{-34} \times 3 \times 10^8}{1.602 \times 10^{-19} \times 1310 \times 10^{-9}}$$

$$\therefore P_{int} = 0.02913 \text{ W} = 29.13 \text{ mW}$$

Q. 2.10 Explain the following terms for semiconductor laser diodes with neat diagram

i) Absorption

ii) Spontaneous Emission

iii) Stimulate Emission.....(FEB-23, MARCH-24, 5 Marks)

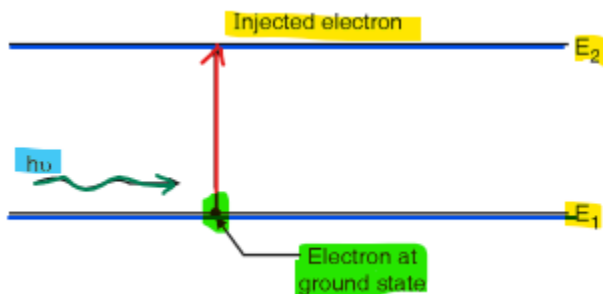
The term LASER stands for light amplification by stimulated emission of radiation. Here the word 'light' stands for electromagnetic radiation which covers the range from the ultraviolet to the infrared region.

The basic steps required to form a laser beam are as follows :

There are generally three processes :

(i) Absorption:

Under normal conditions, the tendency of electrons in the case of the semiconducting material is to stay at the lower energy level. This lower energy level is called the ground state (E_1). If the photon of a certain energy is incident on this electron then the electron absorbs this energy. So the total energy contained by that electron increases. Now that electron can shift toward the higher energy level (E_2). This process is shown in the following figure.



The number of electrons, absorbing the incident photons and traveling toward the higher energy level is more. But for simplicity, only one electron is shown. This transfer of electrons is purely because of an increase in its energy.

An important consideration is that the incident photon should provide sufficient energy to the electron to cause the electron motion from E_1 to E_2 . The sufficient energy in this case is the

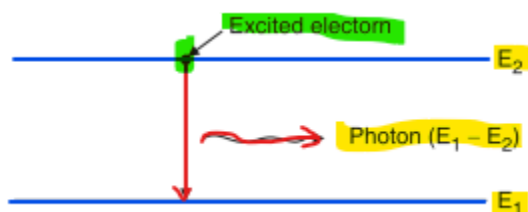
energy level difference between E_1 and E_2 . That means the incident photon should provide at least the energy $E_2 - E_1$. If sufficient energy is not provided by the incident photon, the electron absorbs the energy. But it will not reach the energy level E_2 .

In this case, the electron is shifted and occupies an intermediate energy level between E_1 and E_2 . This intermediate energy level is called a metastable state. After some time the electron falls to its original energy level (the ground state).

(ii) Spontaneous emission:

After the absorption process, the electrons or atoms are present at a higher energy level. That means they possess the excited state. Now if no incident light (that means photons) is provided externally, these electrons stay at this stage for a short time. But after some time, the electrons fall to their original stage which is a lower energy level.

Consider one electron present at the higher energy level. When this electron falls back to the lower energy level, it emits the energy that was absorbed during the absorption process. Thus it gives out the radiation. The energy emitted during this transition of electrons from E_2 to E_1 is simply the energy level difference between these two energy levels. (That means $E_2 - E_1$). These electrons falling back to the ground state may directly come to the ground state or it may follow 2 to 3 steps to reach the ground state. The spontaneous emission process is shown in the following figure.



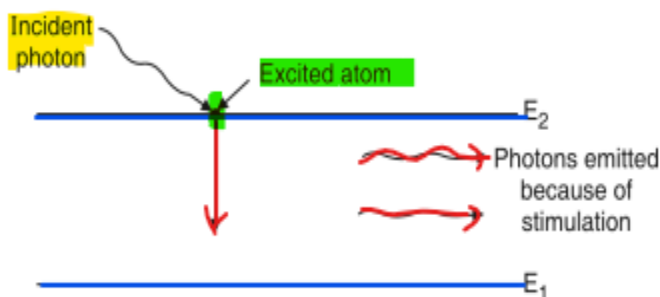
(iii) Stimulated emission:

This is the key process of the formation of a laser beam. Because in this process only the amplification of light takes place. Consider the electron or atom present in the excited state. After a certain time, this atom falls due to a spontaneous process and emits a photon.

But before the occurrence of the spontaneous emission process, if the incident photon is allowed to strike this excited atom then there is no chance for this atom to go to a further higher energy level.

In this case, once the external photon strikes this excited atom; this atom leaves its position from the higher energy level and it will emit the photon. Thus we are getting two photons at the output; one of which is striking on the excited atom and coming out. The other comes out because of the excited atom leaving the higher energy level.

So one input photon causes the emission of two photons and causes light amplification. These two photons are in the same phase and traveling in the same direction. This process is shown in the following figure.



Q. 2.11 List Advantages and disadvantages of Lasers over Other Light Source(5 Marks)

Advantages:

- (1) It has a high information-carrying capacity. So it is used to transmit information in the communication domain.
- (2) It is free from electromagnetic interference. This phenomenon is used in optical wireless communication through free space for telecommunication and computer networking.
- (3) It has very minimum signal leakage.
- (4) Laser-based fiber optic cables are very light in weight and hence are used in fiber optic communication systems.
- (5) It is less damaging compared to X-rays. So it is widely used in the medical field for cancer treatment. It is used to burn small tumors on the eye surface and also on the tissue surface.
- (6) A single laser beam can be focused in areas smaller than 1 micron diameter. One square micro area is needed to store 1bit of data. This helps to store 100 million data in one square cm. Due to this fact, a laser is being used in CDs and DVDs for data storage in the form of audio, video, documents, etc.

Disadvantages:

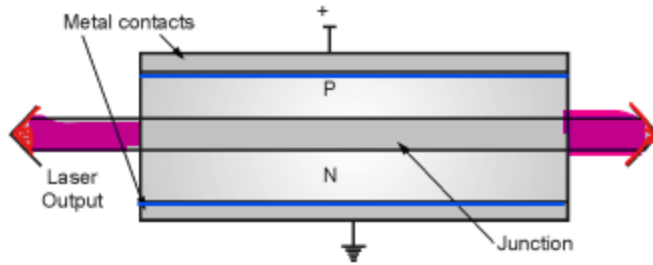
- (1) It is expensive.
- (2) Increases complexity and duration of the treatment based on laser devices or equipment.
- (3) Lasers cannot be used in many commonly performed dental procedures like filling cavities between teeth etc.
- (4) A laser beam is very delicate to handle in the cutting process. A slight mistake in adjusting distance and temperature may lead to burning or discoloring of the metals. Moreover, it requires higher power during the cutting process.

Q. 2.12 Write a short note on semiconductor LASER.....(5 Marks)

The semiconductor laser is a type of laser that is small in appearance and size. The active material used in a semiconductor laser is gallium arsenide and therefore, the laser is also known as Gallium Arsenide Laser.

The working of a semiconductor laser is similar to the PN diode in forward-biased conditions. The PN material is connected to the DC power supply with the help of the metal contacts. The semiconductor laser is also known as the “Injection Laser” because the current is injected into the junction between P and N material.

This consists of a p - n junction made up of Gallium Arsenide material. This junction is forward-biased, so the current starts flowing. It means the electrons start moving from the n side to the p side and the holes start moving from the p side to the n side. Now the recombination of electrons and holes takes place. It means the electrons fall from the conduction band to the valance band. And while doing this they emit the photons. The construction of the semiconductor Laser is shown in the following figure.



Application of Semiconductor Lasers:

This laser is a transmitter of digital data naturally. Because it can be pulsed at different rates and pulse widths.

Advantages of Semiconductor Lasers

- (1) They find many applications due to their small size and appearance.
- (2) These lasers are economical.
- (3) There is no use of mirrors.
- (4) The power consumption is low.

Disadvantages of Semiconductor Lasers

- (1) The beam divergence is more than 125 to 400 milliradians which is greater than other laser types.
- (2) The output beam has an unusual shape as the medium used is short and rectangular.
- (3) The working of this laser type is dependent on the temperature.

Q. 2.13 Compare LED with LASER w.r.t. the following points.

- i) Output Power
- ii) Spectral width
- iii) Coupled Power
- iv) I/O efficiency
- v) Speed

(FEB-23,5 Marks)

PARAMETER	LED	LASER
1)Output Power	Low optical output power	High optical output power
2)Spectral Width	LED has a broader optical spectrum, usually ranging from 25 nm to 100 nm.	For LASER, the optical spectrum is much narrower, usually 0.01 nm to 5 nm.
3)Coupled Power	Low, due to the light spreading	High, due to the focused beam
4)I/O Efficiency	Low conversion efficiency, around 10% to 20%.	High conversion efficiency, around 30% to 70%.
5)Speed	Lower, due to spontaneous emission	Higher, due to stimulated emission

Q. 2.14 A GaAs laser operating at 900nm has a 300μm length and refractive index n=4.3. Find the following
i) Frequency spacing
ii) Wavelength spacing
iii) Number of modes..... (FEB-23,5 Marks)

Given: Wavelength (λ) = 900 nm = 900×10^{-9} m,

Length (L) = 300μm = 300×10^{-6} m,

Refractive index (n) = 4.3

i) Frequency Spacing:

$$\text{Formula} \Rightarrow \Delta f = \frac{c}{2Ln}$$

C = speed of light = 3×10^8 m/sec

$$\therefore \Delta f = \frac{3 \times 10^8}{2 \times 300 \times 10^{-6} \times 4.3}$$

$$\therefore \Delta f = 1.16 \times 10^{11} \text{ Hz}$$

ii) Wavelength Spacing:

Formula $\Rightarrow \Delta\lambda = \frac{\lambda^2}{2Ln}$

$$\therefore \Delta\lambda = \frac{(900 \times 10^{-9})^2}{2 \times 300 \times 10^{-6} \times 4.3}$$

$$\therefore \Delta\lambda = 3.139 \times 10^{-10} \text{ m}$$

$$\therefore \Delta\lambda = 0.3139 \text{ nm}$$

iii) Number of modes:

Formula $\Rightarrow N = \frac{2Ln}{\lambda}$

$$\therefore N = \frac{2 \times 300 \times 10^{-6} \times 4.3}{900 \times 10^{-9}}$$

$$\therefore N = 2866.66 \simeq 2867 \text{ modes}$$

*****END*****